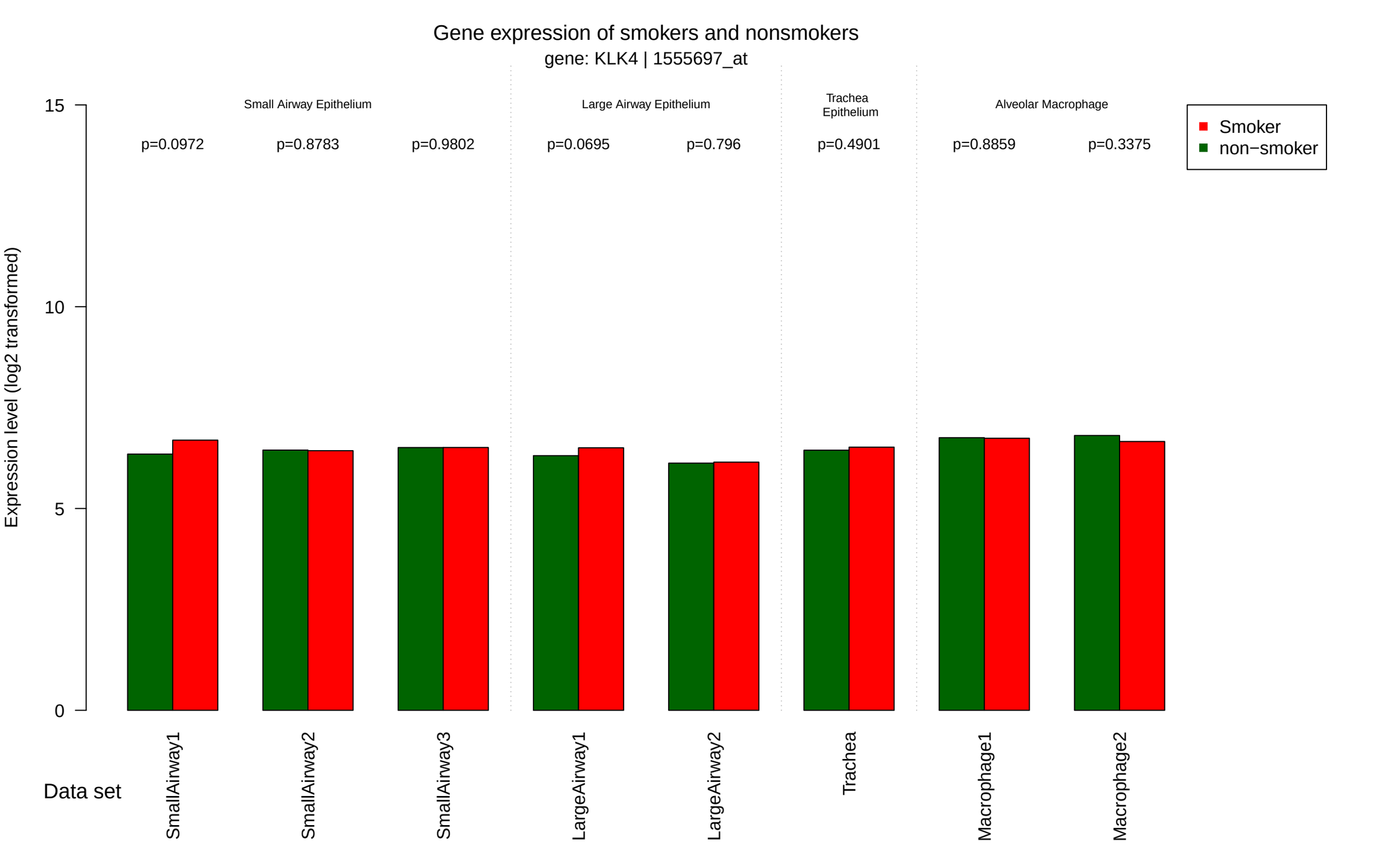
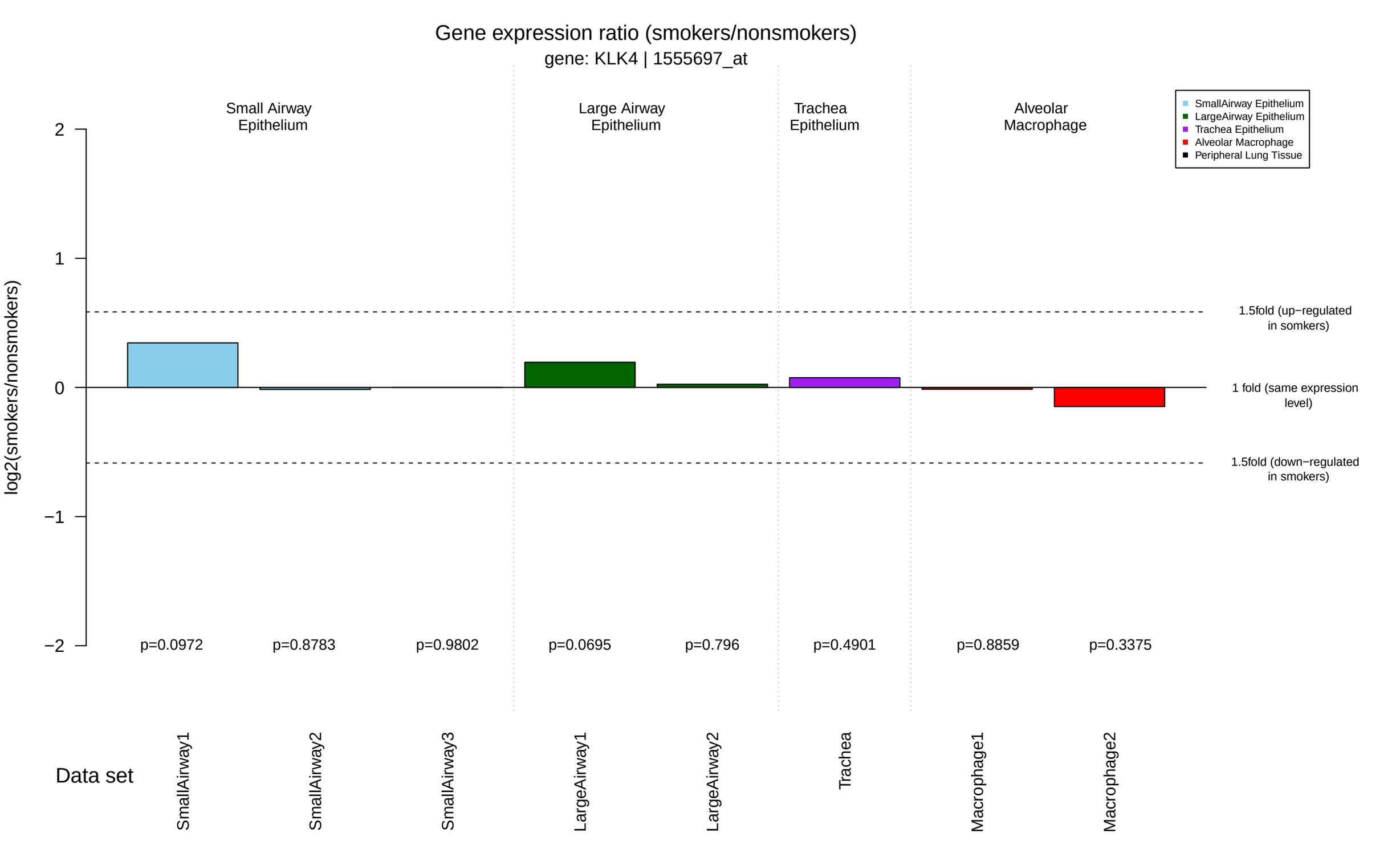
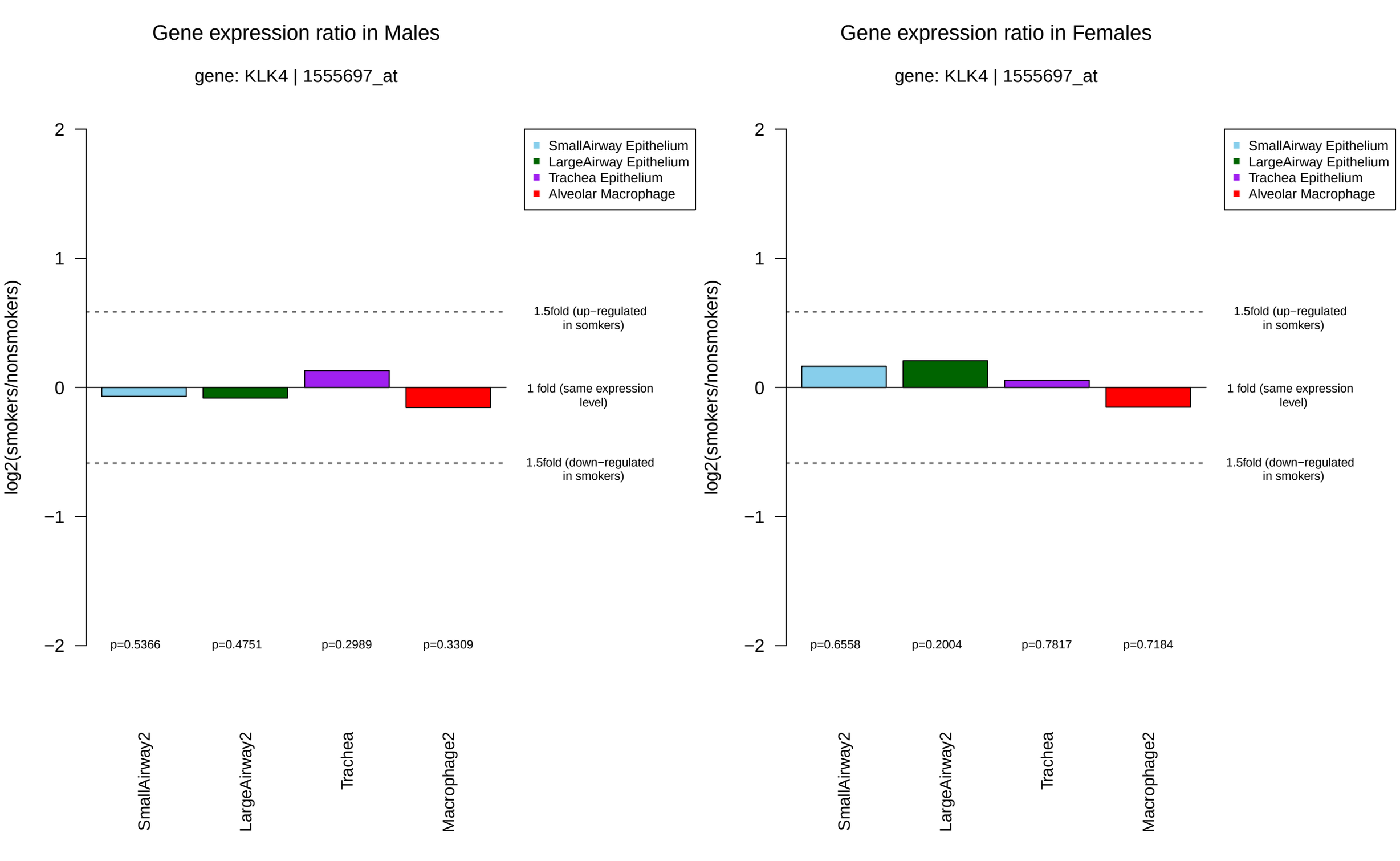
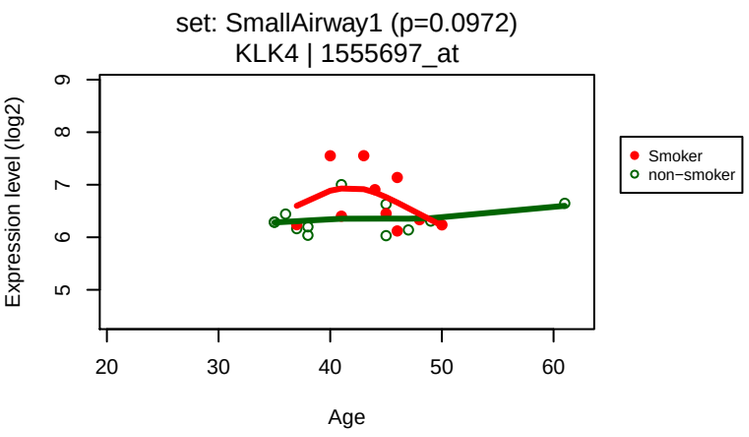
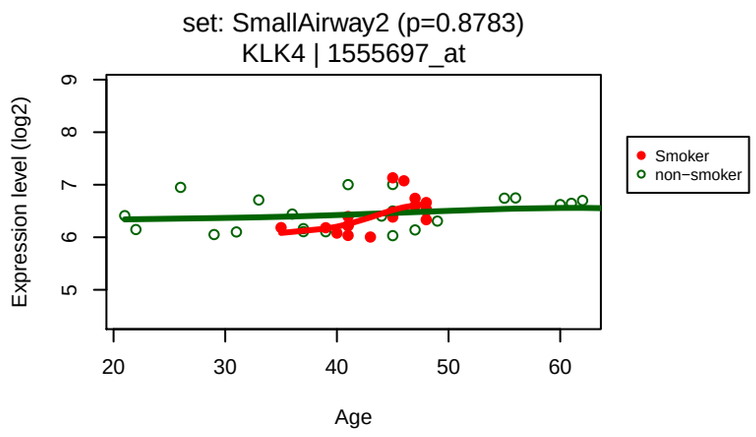
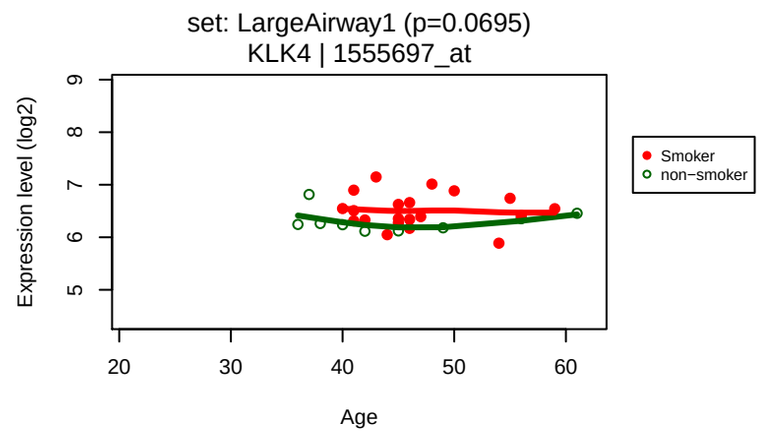

Gene expression of smokers and nonsmokers
gene: KLK4 | 1555697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0972
p=0.8783
p=0.9802
p=0.0695
p=0.796
p=0.4901
p=0.8859
p=0.3375
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

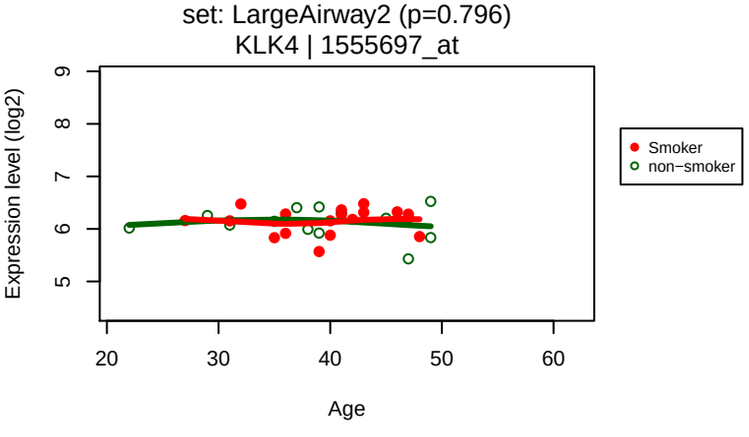

Gene expression ratio (smokers/nonsmokers)
gene: KLK4 | 1555697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0972
p=0.8783
p=0.9802
p=0.0695
p=0.796
p=0.4901
p=0.8859
p=0.3375
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

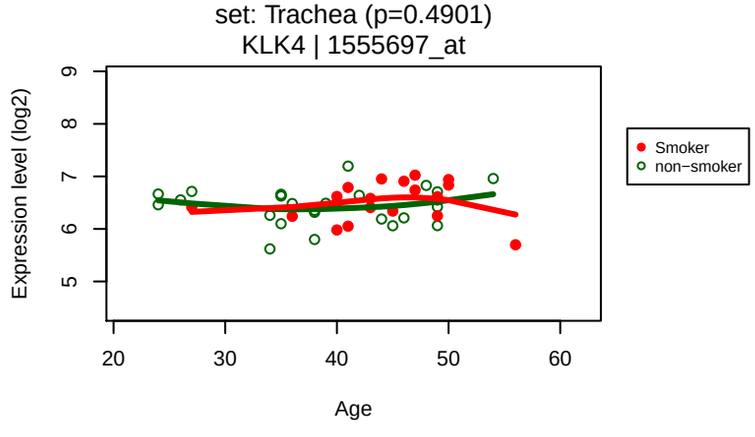

Gene expression ratio in Males
gene: KLK4 | 1555697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5366
p=0.4751
p=0.2989
p=0.3309
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLK4 | 1555697_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6558
p=0.2004
p=0.7817
p=0.7184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

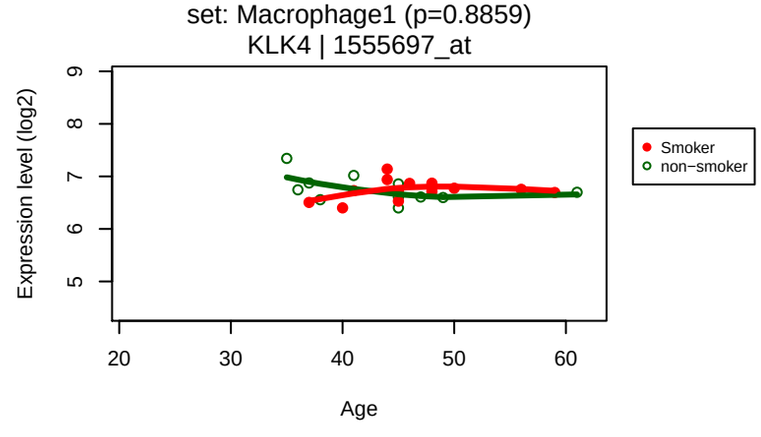

set: SmallAirway1 (p=0.0972)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker

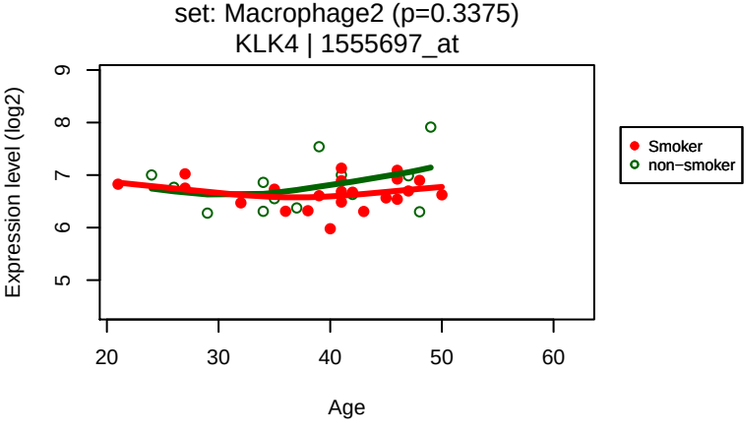

set: SmallAirway2 (p=0.8783)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker

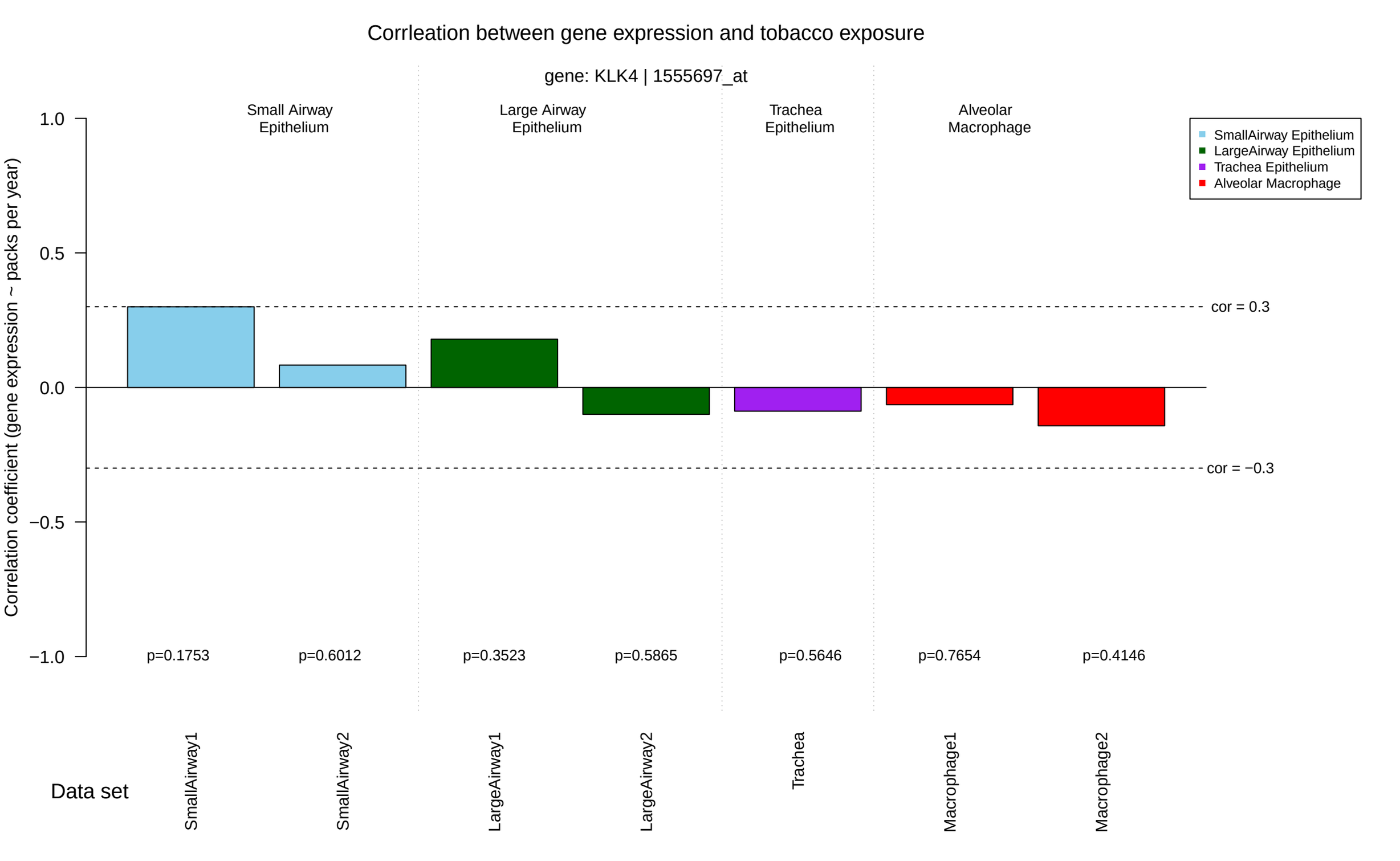

set: LargeAirway1 (p=0.0695)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.796)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4901)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8859)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3375)
KLK4 | 1555697_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLK4 | 1555697_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1753
p=0.6012
p=0.3523
p=0.5865
p=0.5646
p=0.7654
p=0.4146
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2